Kiron Sukesan

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Summary

Software and machine learning engineer with 9+ years of experience spanning ML systems, distributed infrastructure, and biomedical signal processing. Proven track record building large-scale production systems at Amazon and ML platforms at Compass. Research background in Bayesian probabilistic modeling and adaptive signal acquisition with peer-reviewed publications. Seeking senior ML/research engineering roles at the intersection of applied ML and systems.

Skills

Languages	Python, Java, C#, JavaScript, MATLAB, Bash
ML / AI	PyTorch, scikit-learn, Gaussian Process Classification, Bayesian Inference, Active Learning
Systems	Distributed systems, storage node architecture, REST APIs, .NET, LAMP stack
Tools & Infra	AWS (DynamoDB, EC2), Docker, Git, Linux, Arduino, Raspberry Pi, AutoCAD
Domains	ML infrastructure, biomedical signal processing, adaptive sampling, EEG/ECoG analysis

Experience

Software Development Manager *at* **Amazon Web Services (AWS)** Aug 2024 – Present
DynamoDB Storage Node, Seattle, WA

- Lead engineering organization for DynamoDB Storage Node, one of AWS's highest-scale distributed storage systems, overseeing architecture, delivery, and team development.
- Drive cross-functional alignment across engineering, product, and operations for multi-quarter roadmap execution.
- Responsible for hiring, mentoring, and performance management for a team of senior software engineers.

Senior Software Engineer *at* **Amazon Web Services (AWS)** Jan 2023 – Aug 2024
DynamoDB Storage Node API Team, Seattle, WA

- Designed and implemented features for the DynamoDB Storage Node API layer, contributing to a system serving millions of customers at planetary scale.
- Collaborated with principal engineers on storage node architecture to improve reliability, latency, and throughput for core DynamoDB operations.
- Led oncall and operational excellence initiatives to reduce customer-impacting events across the storage tier.

Senior Machine Learning Engineer *at* **Compass** Mar 2021 – Nov 2022
Seattle, WA

- Built and owned ML models powering real estate recommendations and pricing intelligence used by agents across the platform.
- Architected end-to-end ML pipelines from data ingestion through feature engineering, model training, evaluation, and deployment to production.
- Mentored junior ML engineers and drove ML platform improvements to reduce model iteration cycle time.

Machine Learning Engineer II *at* **Compass** Dec 2019 – Mar 2021
Seattle, WA

- Developed and deployed machine learning models for property valuation and market analysis at scale.
- Built data pipelines integrating disparate real estate data sources, improving feature coverage and model accuracy.
- Collaborated with product and data science teams to translate business requirements into ML solutions.

Software Development Engineer II *at* **Amazon** Jul 2019 – Nov 2019
Greater Seattle Area

- Contributed to core Amazon platform services with a focus on reliability, performance, and operational excellence.

Software Development Engineer *at* **Amazon** Oct 2017 – Jul 2019
Greater Seattle Area

- Designed and implemented features for Amazon's internal services, collaborating with senior engineers on distributed systems design and API development.
- Participated in oncall rotations and drove root cause analysis for service reliability improvements.

Data Scientist / Research Engineer *at* **Washington University in St. Louis** May 2016 – Aug 2017
Laboratory of Sensory Neuroscience and Neuroengineering, St. Louis, MO

- Developed a Bayesian active learning framework using Gaussian Process Classification to efficiently estimate audiometric threshold surfaces, reducing clinical measurement burden while preserving diagnostic accuracy.
- Extended the framework to extract latent hearing loss features not measurable by standard audiometry, potentially enabling early-stage detection prior to clinical threshold shifts.
- Implemented a scalable .NET web application for real-time audiogram generation, deployed in ongoing research comparing ML-based audiology methods to standard clinical practice.
- Designed a novel hardware calibration algorithm using ML methods for audiometric equipment.
- Co-authored two peer-reviewed publications in psychometric field estimation and Bayesian audiometry.

Software Developer *at* **Purnatva Solutions Pvt. Ltd.** Sep 2014 – Aug 2015
Bengaluru, India

- Implemented real-time GPS map module for the Falcon Mobile App, deployed to 10,000+ active users.
- Built web features for *vidyadhan.org*, a scholarship matching portal, using the LAMP stack.

Biomedical Engineer (Undergraduate Research) *at* **Washington University in St. Louis** Aug 2013 – May 2014
Department of Biomedical Engineering

- Designed and built a wearable EEG recording device (hardware and software) for overnight clinical use in collaboration with biomedical engineering students and clinicians.

Neurosurgery Research Intern *at* **University of Washington** Jun 2013 – Aug 2013
Brain-Computer Interface Lab (Dr. Jeff Ojemann), Seattle, WA

- Developed a signal analysis algorithm for electrocorticographic (ECoG) neural signals and resolved electromagnetic noise artifacts corrupting recordings.
- Constructed a closed-loop BCI device enabling motor-impaired patient hand movement from theta wave activity.

Education

Washington University in St. Louis 2015 – 2017
Master of Science, Computer Science St. Louis, MO
Washington University in St. Louis 2009 – 2013
Bachelor of Science, Biomedical Engineering St. Louis, MO

Publications

- **Sukesan, K.** et al. "Bayesian active probabilistic classification for psychometric field estimation." *Journal of the Acoustical Society of America* (or relevant venue).
- **Sukesan, K.** et al. "Conjoint psychometric field estimation for bilateral audiometry."